

LUCA GIANFORCARO

Biomechanics Engineer | Robotics and Machine Learning

Date of birth: 14/04/2000

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📍 Turin(TO), Italy

in luca-gianforcaro



EXPERIENCES/PROJECTS

Self Directed Learning

Robot Learning, Perception, Vision

- 📅 August 2026 to Ongoing📍 Remote
- **Coursework:** *Robot Learning: From Fundamentals to Foundation Models* (ETH Zürich, open course materials), theoretical lessons bridging classical control theory and modern multimodal AI; self study of reinforcement learning with Sutton and Barto.
 - **Hands on notebook projects:** neural network implementation in PyTorch, robot control with **LeRobot** and **MuJoCo**, and training of policies with imitation learning and reinforcement learning.
 - **Interested in** perception and vision for robotics, including vision language action models.

Master's Thesis Project

Sant'Anna School of Advanced Studies

- 📅 Sep 2025 to Jul 2026📍 The BioRobotics Institute, Pontedera
- **Master's Thesis:** "Arm Hand Synergic Control for FES Actuation on Hybrid Exoskeleton"
 - Built the data acquisition pipeline: UDP synchronization between the NESM upper limb exoskeleton and the MANUS sensorized glove, and a multimodal dataset of upper limb kinematics from 10 healthy and 7 impaired (emulated with impedance control) subjects across 6 daily living tasks.
 - Built the preprocessing pipeline (synchronization, segmentation, labeling, 300 ms windowing at 100 Hz), producing about 180k windows for time series deep learning models.
 - Designed and trained a BiLSTM-based classifier to decode hand motion intent from proximal arm kinematics: 87.6% median balanced accuracy in leave one subject out validation on healthy subjects.
 - Applied transfer learning from healthy to impaired data to handle domain shift: accuracy raised from about 67% zero shot to 84 to 87.6% after fine tuning.
 - Developed a semi supervised VAE-based model for subject invariant latent representations: 81.8% mean cluster accuracy and 0.787 macro F1.
 - Quantized the classifier with TFLite for the target embedded platform (accuracy difference versus Keras below 0.05%) and started the **ROS2** nodes to drive the stimulation module.

Master Degree Activities

Polytechnic University of Turin

- 📅 2023 to 2026📍 Turin, Italy
- **Multibody in Human Modeling and Biorobotics:** kinematics and dynamics of robotic and musculoskeletal systems (Denavit Hartenberg convention), passive and active elements, control systems (e.g. PID), optimization, Design of Experiments, hardware in the loop. **Project:** humanoid robot in **Adams**; from a provided geometry, modeled contacts and designed the control system.
 - **Applied Biomechanics:** kinematics and dynamics of serial multibody chains, parallel kinematic models of the musculoskeletal system, dynamic systems with damping, free and forced oscillations, transient and impact analysis, random vibrations.
 - **AI in Healthcare (team project):** machine learning to predict the outcome of weaning patients from mechanical ventilation.
 - **Biomedical Signals and Systems:** signal filtering and feature extraction, system modeling and simulation in MATLAB and Simulink.
 - **Molecular Dynamics:** microscale simulations with GROMACS and VMD on Linux.

International Experience

Erasmus+ and study abroad programs

- 📅 2017 to 2022📍 Lisbon, Miami, Dublin
- **Erasmus+ at NOVA School of Science and Technology** (Lisbon, Feb to Jul 2022): coursework in **Signal Theory** and **Biomechanics**, plus Portuguese language and culture course.
 - Study abroad programs: Keiser University (Miami, USA) and ATC school (Dublin, Ireland), 2 weeks each, 2017 to 2018.

EDUCATION

- Master Degree in Biomedical Engineering
Polytechnic University of Turin
- 📅 2023-2026📍 Turin,(TO)
- Mark:110/110 with honor
- Bachelor Degree in Biomedical Engineering
University of Palermo
- 📅 Sep 2019-Mar 2023📍 Palermo,(PA)
- Mark:110/110 with honor
- Scientific High School
High School A.Sciascia
- 📅 Sep 2014-July 2019📍 Canicatti,(AG)
- Mark:96/100

LANGUAGES

- Italian: Native
- English: Certified C1 IELTS
- Portuguese: Certified B1

ACTIVITIES

- **English Course C1 level**
- 📅 2020-2021📍 Wall Street English School
- **Physics and Maths Tutor**
- 📅 2019-2021📍 GoStudent Platform
- **Portuguese Course**
- 📅 2021📍 NOVA FCT of Lisbon, Portugal
- **ROS2 Course**
- 📅 2025📍 Online

SKILLS

Programming & ML

- PythonTensorFlow/KerasPyTorch
- MATLAB/SimulinkLabviewC++ (basic)

Robotics & Systems

- ROS2MuJoCoEmbedded deployment
- UDP real-time communication

Engineering Tools

- SolidworksMimicsPatran-NastranGromacs
- VMDAdams ViewRhinoceros 3DOctave
- Orcad PspiceMSC Adams

INTERESTS

- RoboticsMachine LearningAITechnology
- InnovationSports-FitnessTravelling